# Amey Mishra

Tomball, TX 77375 | 832-422-0725 | amey.mishra2020@gmail.com | ameymishra.com | LinkedIn

## Education

**HARMONY SCHOOL OF ADVANCEMENT** | Houston, TX | Expected May 2027
**Valedictorian | GPA: 4.75 | SAT: 1520 | Mechanical/Aerospace Engineering**
**Advanced Coursework** — Multivariable Calculus, Linear Algebra, College Physics II, AP Calculus BC, AP Research, AP Cybersecurity, AP Environmental Science, AP English Literature.

## Technical Skills

**Mechanical / CAD** — Fusion 360, Onshape, Revit, mechanism design, DFM, 3D printing, prototyping, actuator integration.
**Programming / Controls** — Python, Java, Arduino, OpenCV, sensor feedback, autonomous navigation, path planning, inverse kinematics.

## High-Impact Experience

**FOUNDER, CAPTAIN & MECHANICAL LEAD** | 2023 - Present
**FIRST Tech Challenge Team 25679 The Cosmobots | thecosmobots.com**
- Led robot architecture, mechanical design, fabrication, testing, strategy, and team operations; top worldwide performance among 8,000+ teams, World Championship 4th-ranked alliance, Carolinas Premier Champion, and FTC Hall of Fame.
- FIRST Leadership Award Finalist; reduced a high-cycle transfer mechanism from ~0.75 s to ~0.30 s through iterative CAD, actuator- placement studies, and field testing.

**FOUNDER & CEO** | 2024 - Present
**Cosmobots Robotics | cosmobotsrobotics.com**
- Built a STEM nonprofit supporting four FTC teams and three rocketry teams; secured $50K+ in support, coordinated mentors/budgets/equipment, and built partnerships with Unitree Robotics, Stokes **Education**, Harmony Public Schools, Meiborg, and others.
- Generated 4,500+ organizational service hours and 30+ outreach events; reached 25,000+ people directly and students across seven countries.

**LEAD RESEARCHER & SYSTEMS ENGINEER** | 2025 - Present
**Space Debris Research / NASA Space Apps - ORCA | orcadebriscleanup.space**
- Published research on orbital-debris growth/collision risk; led ORCA, a capture-and-recycling concept using debris-analysis neural networks, orbital mechanics, and CAD/simulation of collection/recycling systems.
- NASA Space Apps Regional Winner and Global Semifinalist; developed mission feasibility, operations, implementation, and business- planning components.

**ROBOTICS & AI RESEARCHER** | 2026 - Present
**Mentored by a Stanford PhD Student**
- Developed a neural-network-powered humanoid arm with custom CAD/3D-printed structure, motors/sensors, adaptive inverse kinematics, Java/Arduino controls, and OpenCV perception.

**DESIGN & MANUFACTURING INTERN** | 2026 - Present
**Pike Robotics**
- Designed pivoting camera mounts and umbilical-management mechanisms in Fusion 360/Onshape for an industrial inspection robot operating inside 100+ ft tanks.

## Additional Engineering & Leadership

**Rice University Research** — Electromagnetic energy-harvesting study; microcontroller test systems, simulation, conference presentation, and research manuscript submission.
**RECF Aerial Drone Competition** — Founder/Captain/Pilot; Python autonomous-flight library; 3 Regional Championships, 3 Excellence/All- Around Awards, 3 Nationals qualifications, Flight Operations Award, Judges Award.
**Rocketry / TVC** — American Rocketry Challenge team lead; designed/tested custom thrust-vector-control system; autonomous flight programs achieved ~97% successful execution in testing.
**Science Olympiad** — President and engineering competitor; 7 Regional Golds, 1 State Gold, 2 State Bronzes in build/engineering events.

## Honors, Projects & Certifications

**Honors** — FIRST Leadership Award Finalist; FTC Hall of Fame / Carolinas Premier Champion; NASA Space Apps Global Semifinalist; Gene Kranz Scholarship; International Math Challenge Gold.
**Projects** — Autonomous mapping UAV; RC aerodynamics test aircraft; humanoid robot; NASA exoplanet CNN.
**Certifications** — Autodesk Fusion 360 & Revit; CompTIA Network+; OpenEDG & Microsoft Python; FAA TRUST; Adobe Photoshop/Illustrator; W3Schools JavaScript.